

JISU JEON

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RESEARCH INTERESTS

Low temperature plasma modeling, Plasma driven solution electrochemistry,
Plasma etching and deposition for microelectronics

EDUCATION

Aug. 2023 – Present

UNIVERSITY OF MICHIGAN

Ann Arbor, USA

Ph.D., Nuclear Engineering and Radiological Sciences

- Computational plasma science and engineering group (Principal Investigator: Mark J. Kushner)
- Developed a 0D liquid chemistry framework coupled to 2D pulsed-plasma reactive gas fluxes through plasma-liquid interaction at the interface and gas species transport into the bulk liquid
- Formulated plasma chemical kinetic mechanisms for applications of PFAS remediation and chain-growth polymerization to provide predictive modelling and optimize experimental configurations
- Relevant courses: Introduction to Low Temperature Plasmas, Plasma Chemistry and Plasma Surface Interactions, Applied Machine Learning (NERS), Machine Learning (EECS, audit)

Sep. 2021 – May. 2023

SOGANG UNIVERSITY

Seoul, Korea

M.S., Mechanical Engineering

- Won the 4th prize at the 7th KSME-SEMES (Korean Society of Mechanical Engineers-Samsung Electronics Mechatronic Solutions) Open Innovation Challenge with a presentation on Monte Carlo collision model considering rotations of molecules to simulate sputtered film deposition
- Relevant courses: Introduction to Nanotechnology and Engineering, Numerical Heat Transfer, Design of Experiments and Statistical Analyses, Classical Physics, Quantum Mechanics

Mar. 2017 – Aug. 2021

SOGANG UNIVERSITY

Seoul, Korea

B.S., Double major in Mechanical Engineering and Physics

- Won the 1st prize at Sogang University Student Academic Conference with a presentation on a simulation of magnetic confinement of electrons in a magnetron sputtering reactor
- Relevant courses: Fluid Mechanics, Continuum Mechanics, Electromagnetism, Heat and Mass Transfer, Finite Element Analysis, Statistical Physics, Modern Optics, Physics Laboratory

PUBLICATION

- (In progress) [Jisu Jeon](#), Mark J. Kushner, Aniline Polymerization in Plasma Driven Solution Electrochemistry; expected to be submitted by Feb. 2027
- (In progress) [Jisu Jeon](#), Tiago C. Dias, et al., A Mechanism for Plasma Induced PFAS-in-Water Degradation on Long-Time Scales; expected to be submitted by Oct. 2026
- Tiago C. Dias, [Jisu Jeon](#), et al., Controlling Reactivity at the Plasma-Liquid Interface: Applications to Remediation of PFAS in Water, Plasma Chemistry and Plasma Processing 46, 84 (2026)

HONORS & AWARDS

- Second prize, Best PhD Oral Presentation by Organizing Committees' Choice at HAKONE XIX International Symposium on High Pressure Low Temperature Plasma Chemistry, 2026
- Fourth prize, The 7th KSME-SEMES Open Innovation Challenge, 2022
- First prize, Sogang University Student Academic Conference, 2020

SKILLS

- Python, Matlab, Fortran, C++, Pointwise, COMSOL Multiphysics®